serviceOrchestrationSimpleStoreManagement

Service Description

Service ID: *"serviceOrchestrationSimpleStoreManagement"*

Abstract
This document describes the `serviceOrchestrationSimpleStoreManagement` service, version 1.0.0, produced by the SimpleStoreServiceOrchestration core system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `serviceOrchestrationSimpleStoreManagement` service, which provides operator management of the orchestration store: the static rules telling which service instance a consumer shall use, with priorities. It is produced by the SimpleStoreServiceOrchestration system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2]. The MQTT binding is specified in the MQTT/TLS/JSON IDD [3].

2.1 Significant Prior Art

Successor of the Arrowhead 4.x Orchestrator store management.

2.2 How This Service Is Meant to Be Used

Operators and support systems such as the PlantDescriptionEngine create rules (consumer, service instance, priority), query, reprioritize and remove them. `pull` then returns the rules of the consumer in priority order.

2.3 Important Delimitations and Dependencies

Rules are not checked against the ServiceRegistry or ConsumerAuthorization at orchestration time.

3 Service Interface

This section describes the interfaces to the `serviceOrchestrationSimpleStoreManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.



Figure 1: SysML-style block of the `serviceOrchestrationSimpleStoreManagement` service and its operations.

The following interface operations are available.

3.1 operation query (OrchestrationSimpleStoreQueryRequest) : OrchestrationSimpleStoreListResponse

Returns a page of store entries matching the filter.

3.2 operation remove (List<String>)

Removes store entries by identifier.

3.3 operation create (OrchestrationSimpleStoreListRequest) : OrchestrationSimpleStoreListResponse

Creates store entries; the service definition is taken from the instance identifier.

3.4 operation modify-priorities (PriorityRequest) : OrchestrationSimpleStoreListResponse

Changes the priorities of store entries.

4 Information Model

Here, all data objects that can be part of the `serviceOrchestrationSimpleStoreManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.8, which are used to represent things like hashes and identifiers.

4.1 struct OrchestrationSimpleStoreListRequest

Store entries to create.

Field	Type	Description
candidates	List<OrchestrationSimpleStoreRequest>	Store entries to create.

4.2 struct OrchestrationSimpleStoreListResponse

List of store entries.

Field	Type	Description
entries	List<OrchestrationSimpleStoreResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.3 struct OrchestrationSimpleStoreQueryRequest

Paged filter for store entries.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
ids	List<Identifier>	Store entry identifiers (UUID) to match.
consumerNames	List<String>	Consumer system names to match.
serviceDefinitions	List<Name>	Service definition names to match.
serviceInstanceIds	List<Identifier>	Service instance identifiers to match.
minPriority	Number	Lower bound of the priority.
maxPriority	Number	Upper bound of the priority.
createdBy	Name	Creator system to match.

4.4 struct OrchestrationSimpleStoreRequest

One orchestration rule: consumer, service instance and priority.

Field	Type	Description
consumer	Name	Consumer system the rule applies to.
serviceInstanceId	Identifier	Service instance offered to the consumer.
priority	Number	Priority of the rule; 1 is the highest.

4.5 struct OrchestrationSimpleStoreResponse

A stored orchestration rule.

Field	Type	Description
id	Identifier	Identifier (UUID) of the store entry.
consumer	Name	Name of the consumer system.
serviceDefinition	Name	Service definition name.
serviceInstanceId	Identifier	Service instance identifier `<System>|<serviceDefinition>|<version>`.
priority	Number	Priority of the entry; 1 is the highest.
createdBy	Name	Name of the system that created the record.
updatedBy	Name	Name of the system that changed the record last.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.6 struct Page

Paging and sorting; all fields optional.

Field	Type	Description
page	Number	Zero-based page number.
size	Number	Number of records per page (at most `max.page.size`, default 1000).
direction	String	Sort direction, `ASC` or `DESC`.
sortField	String	Field to sort by.

4.7 struct PriorityRequest

A JSON object mapping store entry identifiers (UUID) to new priorities, e.g. `{"123e4567-e89b-12d3-a456-426614174000": 5}`.

4.8 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.

5 References

[1] J. Delsing, “SimpleStoreServiceOrchestration – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, serviceorchestration-simple/documentation/v5.3.0/SimpleStoreServiceOrchestration-v5.3.0-SysD.pdf.

[2] ——, “serviceOrchestrationSimpleStoreManagement HTTP/TLS/JSON – Interface Design Description (SimpleStoreServiceOrchestration),” Eclipse Arrowhead, IDD 5.3.0, 2026, simpleStoreServiceOrchestration-v5.3.0-serviceOrchestrationSimpleStoreManagement-HTTP-IDD.pdf.

[3] ——, “serviceOrchestrationSimpleStoreManagement MQTT/TLS/JSON – Interface Design Description (SimpleStoreServiceOrchestration),” Eclipse Arrowhead, IDD 5.3.0, 2026, simpleStoreServiceOrchestration-v5.3.0-serviceOrchestrationSimpleStoreManagement-MQTT-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	